

SELLER AND INVENTORY MANAGEMENT

Pepperfry – E-Commerce Database Management System

1. Database Design

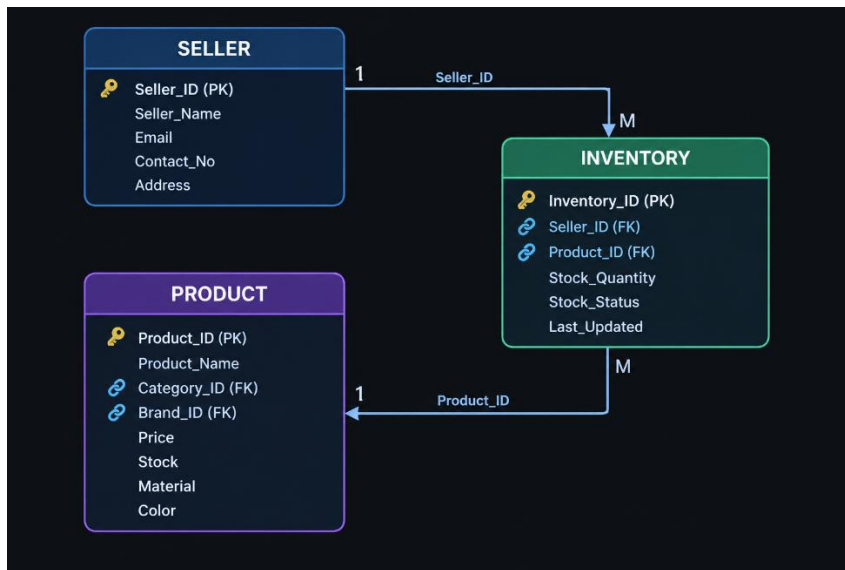
1.1 Seller Table

Attribute	Data Type	Constraint
Seller_ID	NUMBER	Primary Key
Seller_Name	VARCHAR2(100)	NOT NULL
Email	VARCHAR2(100)	NOT NULL, UNIQUE
Contact_No	VARCHAR2(15)	NOT NULL, UNIQUE
Address	VARCHAR2(200)	NOT NULL

1.2 Inventory Table

Attribute	Data Type	Constraint
Inventory_ID	NUMBER	Primary Key
Product_ID	NUMBER	Foreign Key
Seller_ID	NUMBER	Foreign Key
Stock_Quantity	NUMBER	NOT NULL
Stock_Status	VARCHAR2(20)	NOT NULL
Last_Updated	DATE	NOT NULL

2. Table Relationships



3. Table Creation

3.1 Create Seller Table

```
SQL> CREATE TABLE Seller (  
2     Seller_ID NUMBER PRIMARY KEY,  
3     Seller_Name VARCHAR2(100) NOT NULL,  
4     Email VARCHAR2(100) UNIQUE,  
5     Phone VARCHAR2(15),  
6     Address VARCHAR2(200)  
7 );
```

Table created.

3.2 Create Inventory Table

```
SQL> CREATE TABLE Inventory (  
2     Inventory_ID NUMBER PRIMARY KEY,  
3     Product_ID NUMBER,  
4     Seller_ID NUMBER,  
5     Stock_Quantity NUMBER NOT NULL,  
6     Stock_Status VARCHAR2(20) NOT NULL,  
7     Last_Updated DATE NOT NULL,  
8     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID),  
9     FOREIGN KEY (Seller_ID) REFERENCES Seller(Seller_ID)  
10 );
```

Table created.

4 . Seller Data Insertion

```

SQL> INSERT INTO Seller VALUES
  2 (101, 'Urban Furniture', 'urban@gmail.com', '9876543210', 'Chennai');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (102, 'Home Style', 'homestyle@gmail.com', '9876543211', 'Bangalore');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (103, 'Wood Craft', 'woodcraft@gmail.com', '9876543212', 'Coimbatore');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (104, 'Royal Furnishings', 'royal@gmail.com', '9876543213', 'Madurai');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (105, 'Comfort Homes', 'comfort@gmail.com', '9876543214', 'Hyderabad');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (106, 'Modern Living', 'modern@gmail.com', '9876543215', 'Mumbai');

```

5. Inventory Data Insertion

```

SQL> INSERT INTO Inventory VALUES (1, 1, 101, 50, 'In Stock', SYSDATE);

1 row created.

SQL> INSERT INTO Inventory VALUES (2, 2, 102, 35, 'In Stock', SYSDATE);

1 row created.

SQL> INSERT INTO Inventory VALUES (3, 3, 103, 25, 'In Stock', SYSDATE);

1 row created.

SQL> INSERT INTO Inventory VALUES (4, 4, 104, 40, 'In Stock', SYSDATE);

1 row created.

SQL> INSERT INTO Inventory VALUES (5, 5, 105, 20, 'Low Stock', SYSDATE);

1 row created.

SQL> INSERT INTO Inventory VALUES (6, 6, 106, 60, 'In Stock', SYSDATE);

1 row created.

SQL> INSERT INTO Inventory VALUES (7, 7, 107, 30, 'In Stock', SYSDATE);

1 row created.

SQL> INSERT INTO Inventory VALUES (8, 8, 108, 45, 'In Stock', SYSDATE);

1 row created.

```

6. Seller Product Information

6.1 Display Seller and Product Details

```

SQL> SELECT
2     s.Seller_ID,
3     s.Seller_Name,
4     p.Product_ID,
5     p.Product_Name,
6     p.Price,
7     p.Material,
8     p.Color
9 FROM Seller s
10 JOIN Inventory i
11     ON s.Seller_ID = i.Seller_ID
12 JOIN Product p
13     ON i.Product_ID = p.Product_ID;

```

```

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
PRICE MATERIAL
-----
COLOR
-----
101
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
PRICE MATERIAL
-----
COLOR
-----
Urban Furniture
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----

```

7. Inventory Status

7.1 Display Available Products

```
SQL> SELECT
2      p.Product_ID,
3      p.Product_Name,
4      i.Quantity
5  FROM Product p
6  JOIN Inventory i
7  ON p.Product_ID = i.Product_ID
8  WHERE i.Quantity > 0;
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY
-----
1
Sofa      50

2
Dining Table 40

PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY
-----
3
Office Chair 25

4
Bed

PRODUCT_ID
```

7.2 Display Unavailable Product

```
SQL> SELECT
2      p.Product_ID,
3      p.Product_Name,
4      i.Quantity
5  FROM Product p
6  JOIN Inventory i
7  ON p.Product_ID = i.Product_ID
8  WHERE i.Quantity = 0;
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY
-----
2
Dining Table 0
```

8.Inventory Update

```

SQL> UPDATE Inventory
2 SET Quantity = 0,
3     Reorder_Level = 5,
4     Last_Updated = SYSDATE
5 WHERE Product_ID = 2;

1 row updated.

SQL>
SQL> COMMIT;

Commit complete.

SQL> SELECT *
2 FROM Inventory
3 ORDER BY Inventory_ID;

```

INVENTORY_ID	PRODUCT_ID	SELLER_ID	QUANTITY	REORDER_LEVEL	LAST_UPDA
1	1	101	50	5	17-SEP-26
2	2	102	0	5	17-SEP-26
3	3	103	25	5	17-SEP-26
4	4	104	35	5	17-SEP-26
5	5	105	20	5	17-SEP-26
6	6	106	45	5	17-SEP-26
7	7	107	30	5	17-SEP-26
8	8	108	15	5	17-SEP-26
9	9	109	50	5	17-SEP-26
10	10	110	25	5	17-SEP-26

```

10 rows selected.

```

9. Inventory Status Reports

9.1 Complete Inventory Report

```

SQL> SELECT
2   i.Inventory_ID,
3   p.Product_ID,
4   p.Product_Name,
5   s.Seller_ID,
6   s.Seller_Name,
7   i.Quantity,
8   i.Reorder_Level,
9   CASE
10      WHEN i.Quantity = 0 THEN 'Unavailable'
11      WHEN i.Quantity <= i.Reorder_Level THEN 'Low Stock'
12      ELSE 'Available'
13   END AS Stock_Status,
14   i.Last_Updated
15 FROM Inventory i
16 JOIN Product p
17   ON i.Product_ID = p.Product_ID
18 JOIN Seller s
19   ON i.Seller_ID = s.Seller_ID
20 ORDER BY i.Inventory_ID;

```

INVENTORY_ID	PRODUCT_ID	PRODUCT_NAME	SELLER_ID	SELLER NAME	QUANTITY	REORDER_LEVEL	STOCK_STATU	LAST_UPDA
1	1				50	5	Available	17-SEP-26
Sofa								
101								
Urban Furniture								
50					5	5	Available	17-SEP-26

9.2 Available and Unavailable Product Count

```
SQL> SELECT
  2     SUM(CASE WHEN Quantity > 0 THEN 1 ELSE 0 END) AS Available_Product_Count,
  3     SUM(CASE WHEN Quantity = 0 THEN 1 ELSE 0 END) AS Unavailable_Product_Count
  4 FROM Inventory;
```

AVAILABLE_PRODUCT_COUNT	UNAVAILABLE_PRODUCT_COUNT
9	1

9.3 Total Stock by Seller

```
SQL> SELECT
  2     s.Seller_ID,
  3     s.Seller_Name,
  4     SUM(i.Quantity) AS Total_Stock
  5 FROM Seller s
  6 JOIN Inventory i
  7   ON s.Seller_ID = i.Seller_ID
  8 GROUP BY
  9     s.Seller_ID,
 10     s.Seller_Name
 11 ORDER BY
 12     s.Seller_ID;
```

SELLER_ID

SELLER_NAME

TOTAL_STOCK

101
Urban Furniture
50

102
Home Style
0

SELLER_ID

SELLER_NAME

TOTAL_STOCK

103
Wood Craft
25

104
Royal Furnishings

SELLER_ID

SELLER_NAME

TOTAL_STOCK

35

9.4 Products with Zero Stock

```
SQL> SELECT
  2     s.Seller_ID,
  3     s.Seller_Name,
  4     SUM(i.Quantity) AS Total_Stock
  5 FROM Seller s
  6 JOIN Inventory i
  7     ON s.Seller_ID = i.Seller_ID
  8 GROUP BY
  9     s.Seller_ID,
 10     s.Seller_Name
 11 ORDER BY
 12     s.Seller_ID;
```

SELLER_ID

SELLER_NAME

TOTAL_STOCK

101
Urban Furniture
50

102
Home Style
0

SELLER_ID

SELLER_NAME

TOTAL_STOCK

103
Wood Craft
25

104
Royal Furnishings

SELLER_ID

SELLER_NAME

TOTAL_STOCK

35